

The fact is, NTFS allows the same file to exist in multiple folders at the same time, or under different names in the same folder! We are not talking about copies here, to be clear; NTFS allows for multiple files in multiple directories to point to the same data on the hard disk. This way the data is stored once, and each “copy” doesn’t take more space.

Unfortunately Windows doesn’t include good tools to actually make use of this functionality. So we recommend Link Shell Extension [link / path here] a free utility that add support for creating links to Windows Explorer. With it you can work the following:

- **Hardlinks:** Hard links are what we have described above. Multiple file names and locations all pointing to the same piece of data on the hard drive. If you create a hardlink to a file, it will start to exist in another location, but you can edit / view the file from either location. A hard link to a file has to be created in the same drive as the original file.
- **Symbolic Links:** These are more like shortcuts, is that you can create a link to a file in another drive or even on the network. Unlike shortcuts, symbolic links are transparent to applications, they will open the file directly instead of first interpreting the link.
- **Junctions:** Junctions let you create a link to a folder like you create a symbolic link. So a folder such as D:\folder1\folder2 could be linked to C:\foldera\folderb. You could access all files of D:\folder1\folder2 could be accessed C:\foldera\folderb. Any operations you perform on the linked location, such as adding / deleting files will actually happen in the original folder.

- **Volume mountpoints:** If you’ve used Linux, you’ll know that it has a very different way of handling partitions, instead of C:, D:, E: etc, in Linux partitions are mounted as “folders”, so / harddrive, /home, /media/cdrom etc. Something similar is possible with Windows! You can have a new partition show up as C:\partition instead of E:

Take advantage of fsutil

This is a nice utility included with Windows XP and above that allows you to control many low-level features of NTFS. NTFS is an advanced filesystem with many features, but this also means that unused features can impact performance. Some of these can be turned off or turned on for better performance.

Fsutil: Disable old DOS names

In the old DOS days, file names were limited to 8 characters, and each file could have a 3 character extension at most. A lot has changed since then, and now file names can be as long as 255 characters of Unicode — so you can even have file names in Hindi.

Windows 95 and 98 relied extensively on DOS, and thus they needed to maintain the old 8.3 names in addition to the longer file names. To maintain backward compatibility with older OSs even Windows 7 generates these file names. If you don’t use any obscenely old software, you are better off turning this feature off.

To disable creating these file names, run the following command using the run dialog or the commandline:

```
fsutil 8dot3name set 1
```

This will disable it for all drives, while the following will disable it for all drives except the Windows system drive:

```
fsutil 8dot3name set 3
```

Another way to do this — use this if you are running Windows XP — to run the following command:

```
fsutil behavior set disable8dot3 1
```

Fsutil: Disable Last Access timestamp

Windows maintains a Last Access timestamp for each file and directory that keeps a relatively up to date record of when the file / directory was last accessed / modified. The trouble is, this will be updated each time you even open a folder. The latest timestamp will be written to disk, and this can have a performance impact especially if there is an application making a lot of updates to the disk.

However, this is a core feature and some applications rely on this information. According to Microsoft’s own documentation, their Backup and Remote Storage applications / services need this enabled to function. Be caution and use only if necessary, there may be other third-party applications that need this information.

You can disable this timestamp using the following command:

```
fsutil behavior set disablelastaccess 1
```

Fsutil: Encrypt the paging file

You RAM has some of your most personal data, since it any running, or even non running applications will have stored some stuff there. This would include private documents, passwords etc. Once you switch your computer off though, all that data is lost so there isn’t much to worry about.

However your computer also uses your paging file to store such data, and that paging file survives reboots. Yes, there is an option to delete the paging file at each boot, but that is not

enough, since deleted files can easily be recovered. This feature will make your private data more secure, but it also has a performance impact, so be wary.

```
fsutil behavior set encryptpagingfile 1
```

Changes will occur on reboot.

Pick the right Allocation Unit Size

You may have noticed an option to select an Allocation Unit Size while formatting a drive, and possibly wondered what it does. This setting can have an impact on performance and storage efficiency so it is important to understand how to select it and why.

The allocation unit size also known as the cluster size. A cluster is the basic unit of storage on the disk. Your file size might be measured in bytes, kilobytes etc, but it is stored in clusters. What this means is, if you have a 4 kB cluster size, the minimum amount of space that will be allocated for a file, will be 4 kB. So a file whether it is bit or the full 4 kB will take 4 kB of space; additionally a 6kB file will take 8 kB, i.e. two clusters. This is why there are two values for file / folder size, “Size” and “Size on Disk” in File properties. This difference can be quite significant — just check a folder with many files — since the differences add up.

For maximum efficiency it is best that the average file size be a multiple of the cluster size. This way lesser space is wasted per file. Since you can only change this value while formatting a partition, it isn’t possible to know this in advance. However you can estimate this and accordingly organize your files. If you are storing larger files, such as music, videos etc. on a partition, use a higher cluster size, know that if you pick a size larger than 4 kB, then you will not be able to use